

Phase 3.2 — Next-Step Working Note

v32.15: Robustness and Generalization of the Degeneracy-Closed Interference Law

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Purpose of the next step

Phase 3.2 has now reached a point where the local hotspot mechanism is no longer represented only by an empirical scalar descriptor. The analysis has produced a basis-invariant, degeneracy-closed formulation of the virtual $P \rightarrow Q \rightarrow P$ channels,

$$T_G(E) = g_\eta(E - E_G) PVP_GVP, \quad (1)$$

with

$$\Sigma_E = \sum_G T_G(E). \quad (2)$$

The corresponding normalized interference parameter is

$$A_E^{(G)} = \frac{2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K)}{\left(\sum_G \|T_G\|_F \right)^2}. \quad (3)$$

In v32.14, this basis-invariant quantity remained positively associated with hotspot prominence for both perturbation families. The next scientific question is therefore no longer whether an interference mechanism exists in the present frozen model. The question is whether the same mechanism survives reasonable changes to the model.

Next-step question: Is the degeneracy-closed interference law a robust structural feature, or is it specific to the present 8-qubit, 11-term realization?

What v32.15 should test

The purpose of v32.15 is a controlled robustness and generalization study. Only one model ingredient should be changed at a time so that any loss or persistence of the signal can be interpreted physically.

The primary variations should be:

- **Hamiltonian term count:** repeat the analysis for nearby frozen term counts, for example 7, 9 and 11 terms.
- **Hamiltonian seeds:** increase the number of independent seeds beyond the current set of 12.
- **Regularization scale:** vary η over a narrow, physically reasonable range around the current value, without retuning it to optimize the result.
- **Perturbation families:** retain dephasing and transverse families as separate tests and require the physical direction of the effect to remain consistent.

The main observables should remain the same:

$$A_E^{(G)} \quad \text{and} \quad C_E^{(G)} = \frac{\|\sum_G T_G\|_F}{\sum_G \|T_G\|_F}. \quad (4)$$

No broad feature search should be introduced in this step. The purpose is confirmation, not discovery of a new descriptor.

Decision criteria

A successful robustness result should satisfy three levels of evidence.

Directional stability

The correlation between hotspot prominence and the interference/coherence measure should retain the same sign across most seeds and model variants.

Cross-seed persistence

The median seedwise rank correlation should remain materially above zero when the Hamiltonian term count, seed ensemble or η is changed.

No narrow tuning requirement

The effect should not depend on a single exceptional term count, seed or regularization value. A moderate reduction in effect size is acceptable; a sign reversal or complete disappearance under small model changes would indicate model specificity.

Interpretation of possible outcomes

If the interference signal survives the robustness scan, the Phase 3.2 result can be promoted from a model-specific mechanism candidate to a more general structural statement:

hotspot prominence increases with constructive, degeneracy-closed $P \rightarrow Q \rightarrow P$ interference	(5)
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within the class of Hamiltonians tested.

If the signal weakens strongly or disappears, that is also scientifically useful. It would identify which ingredients of the frozen model are essential for the hotspot mechanism and would prevent an unjustified claim of universality.

Expected role of v32.15 in Phase 3.2

v32.15 should therefore function as a **robustness gate**. It is not intended to create another fitted law. Its role is to determine whether the basis-invariant mechanism established in v32.14 is stable enough to support the closing theory section of Phase 3.2.

A positive result would justify moving next to formal theory writing and an analytic interpretation of the interference law. A negative result would instead define the boundary conditions under which the mechanism exists.

Working status: The current interference law is supported within the frozen reference ensemble. v32.15 is designed to test generalization beyond that ensemble.